# KEVIN WILLIAMS

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## TECHNICAL SKILLS

**Systems Engineering:** Requirements Development & Verification, V&V Planning, MBSE, Systems Lifecycle (NPR 7123, NPR 7150.2, NPR 8739), Interface Management, Milestone Reviews

**AI / ML:** PyTorch, TensorFlow, Hugging Face (Transformers, Spaces), LangChain, CLIP, FAISS, Opacus; LLM integration (RAG, chatbased CRUD), CNNs, ensemble methods, differential privacy, adversarial robustness, red-teaming

**Organizational Design & Leadership:** Organizational design and operating model development, governance frameworks, quality and compliance programs, policy and documentation standards, personnel qualification standards

**Software:** Python, SQL, JavaScript, Django, Flutter, FastAPI, Docker, MySQL, Git, AWS

**Materials & Test Engineering:** Metallography, Microscopy, DFMEA/PFMEA, Black Belt Six Sigma

## EXPERIENCE

**Systems Engineer / Software Architect — Barrios Technology (NASA Johnson Space Center)** **Jan 2026 – Present**

*Houston, TX*

- Engineer designing and building the EHP Program Hub Dashboard, cataloging 100+ tools across 9 offices to consolidate tool visibility for the Extravehicular Activity and Human Surface Mobility Program, with planned extension to CLDP, ISS, and other programs.
- Delivered the dashboard as a vanilla HTML/CSS/JS single-file SPA per deliberate audit-friendly deployment constraints, partnering with IT and engineering leads across 9 offices to define schema and drive structured data collection.

**Software Engineer — Barrios Technology** **Feb 2025 – Dec 2025**

*Houston, TX*

- Architected and built Cortex, an enterprise data platform with integrated LLM capabilities (chat-based CRUD, inline suggestions, data extraction) for Intuitive Machines, built on Django with an EAV architecture and MySQL backend.
- Deployed Cortex internally at Barrios for dev/test validation; delivered working enterprise platform to customer-readiness stage.

**Systems Engineer — Barrios Technology (NASA Johnson Space Center)** **Oct 2023 – Feb 2025**

*Houston, TX*

***ISS Commercial Element — Axiom Space Integration*** *Oct 2023 – Feb 2024*

- Traced requirements for SSP 51074 CERD covering Axiom Space docking integration with ISS, advancing document production and release through NPR 7123 milestone reviews including SRR.

***Orion Multi-Purpose Crew Vehicle (Artemis 1 → Artemis 2)*** *Feb 2024 – Jun 2024*

- Evaluated Flight Test Objectives (FTOs) across 18 subsystems, including aerodynamics, thermal protection, propulsion, GNC, ECLS, avionics, and software, to deliver the Artemis I Flight Analysis Report at 90% completion.
- Integrated inputs from Lockheed Martin and NASA subsystem leads across 18 subsystems, implementing flight analysis data management in SharePoint-based program data systems.

***Commercial LEO Development Program (CLDP)*** *Jun 2024 – Feb 2025*

- Served as NASA SE&I reviewer across four CLDP commercial partners, evaluating physics-based subsystem, ECLSS, structural, HITL, and SRR packages spanning four independent station architectures. Conducted ~40 reviews in 9 months, including CDPO-delegated lead review authority for Blue Origin, and generated RFAs and RIDs reported directly to the Program Office. Drove partner-implemented design changes that tightened technical baselines and resolved integration risks ahead of downstream milestones.
- Led requirements traceability for CLDP, integrating partner flight and ground system requirements against a fragmented top-level baseline spanning three commercial providers. Traced 300+ requirements to CLDP-REQ-1130 and generated RIDs resolving gaps, conflicts, and ambiguities across partner submissions. Delivered a unified requirements baseline adopted by the CLDP Program Office, NASA, and all three partners.
- Developed an AI requirements framework to address an emerging area in commercial human spaceflight, authoring 70 requirements by building consensus across SMEs from four organizations (University of Michigan, FAA, ISO, and NASA) before handing off to NASA for vetting, and continue contributing through cofounded AI-in-Flight working group.

**Engineering Specialist — Ford Motor Company, Van Dyke Electric Powertrain Center** **Jun 2022 – Sep 2023**

*Van Dyke, MI*

- Led launch and served as deputy manager for a 20-person eMotor materials laboratory (12 hourly, 8 salary) on 24/7 shift coverage, testing stators and rotors at hundreds of parts per week for F-150 Lightning and Maverick Hybrid programs.
- Owned full lab buildup and personnel operations — equipment acquisition, hiring support and interview participation, 24/7 shift scheduling, and timecard approvals for hourly staff — alongside test procedure development, calibration, and certification. Achieved ISO certification, internal Ford standards qualification, and production readiness.
- Self-initiated a deep learning computer vision pipeline (PyTorch regression with Kalman filtering for segmentation) for varnish surface area measurement on eMotor stators, replacing manual operator workflow and achieving 99% accuracy with 1+ hour reduction per quality check. Recognized with Ford Technical Excellence Award, June 2023.
- Conducted failure analysis and supplier investigations on eMotor components, driving DFMEA/PFMEA design iterations to resolve material degradation and manufacturing defects.

**Undergraduate Research Assistant — University of Michigan** **Sep 2019 – Jun 2022**

*Ann Arbor, MI | Advisor: Prof. Mark Meyerhoff*

- Developed (as first author) a novel electrochemical detection method for Δ9-THC on disposable screen-printed carbon electrodes (1–20 µM range, 0.13 µM limit of detection, $R^2$ = 0.995) — third-lowest LOD among comparable SPCE devices at time of publication.
- Published peer-reviewed findings (2023) demonstrating viability for in-field law enforcement detection of Δ9-THC in saliva.

**Early Career — Materials & Chemical Engineering** **2012 – 2017**

- **General Motors, Flint Engine** (2015–2016): Established the metallurgical lab supporting the 2017 SGE crankshaft line; performed metallurgical analysis and material validation.
- **Hyundai-Kia** (2017): Led supplier corrective actions, material validation, and quality testing.
- **Quaker Chemical** (2012–2015, part-time): Delivered QA protocol training for personnel and conducted metallurgical analysis and material validation.

## RESEARCH PROJECTS — M.S. AI

**µStruct — AI-Powered Microstructure Identification** *ASTM Project*

- Built ensemble classifier combining CLIP zero-shot inference with FAISS similarity-weighted voting (40/60 split) across 6,000+ micrographs spanning 16 steel microstructural phases.
- Designed and validated grading accuracy methodology against ground-truth phase labels; productionized as FastAPI backend with Docker deployment on Hugging Face Spaces.
- Exposed REST API for retrieval, classification, and evaluation. Live demo: kevwill-microstructure-classifier.hf.space

**Privacy-Preserving Cognitive Twin — Membership Inference Attacks & Defenses** *CIS 545*

- Designed adversarial evaluation pipeline covering confidence-based, label-only, and shadow-model membership inference attack variants.
- Implemented and benchmarked three defenses — DP-SGD via Opacus, Model Confidence Exclusion, and a fusion defense — with ε-value sweeps mapping privacy-utility tradeoffs across attack success rate, AUROC, accuracy, and calibration. **Disaster Infrastructure Damage Recognition via Computer Vision** *CIS 583*
- Developed CNN-based binary classifier (disaster / non-disaster) for infrastructure damage assessment in post-event aerial and ground imagery.
- Evaluated model performance to establish classification confidence thresholds for first-responder deployment readiness.

## EDUCATION

**M.S. Artificial Intelligence** — University of Michigan | GPA 3.8/4.0 **May 2026**

**B.S. Mechanical Engineering** — University of Michigan **December 2021**

## PUBLICATIONS

Williams, K. *Requirements and Verification Standards for Artificial Intelligence in Safety-Critical Applications.* Safety Critical Labs. Zenodo, 2026. DOI: 10.5281/zenodo.19323435

Williams, K. J., White, C. J., Hunt, A. P., & Meyerhoff, M. E. *Differential pulsed voltammetry of Δ9-Tetrahydrocannabinol (THC) on disposable screen-printed carbon electrodes: A potential in-field method to detect Δ9-THC in saliva. Electroanalysis*, 2023. DOI: 10.1002/elan.202200451

## AWARDS

- **Silver Bear Award** (HSFTIC, July 2025) — Leading role in integrating AI into NASA's Software Engineering Handbook; influenced proposed updates to software engineering standards and positioned Barrios and CLDP at the forefront of AI innovation in human spaceflight.
- **Ford Technical Excellence Award** (June 2023) — Implementation of deep learning model technology for quality decisions on eMotor stator varnish inspection.